

Computer Systems 214

Number systems

Lecture 2

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Lecture Overview

- What is a number system?
- Decimal numbers *0 - 9 digits*
- Binary numbers
- Hexadecimal numbers
- Octal numbers
- Conversion between number systems
- Bytes, Nibbles, LSB, MSB, etc.
- Representing letters (ASCII table)

Relevant section in textbook: 1.4.1-1.4.4

Number systems intro

- Refers to the numerical representation of a value.
- A good number system will:
 - Represent a useful set of numbers (e.g. integers, or rational numbers)
 - Give every number represented a unique representation
 - Reflect the algebraic and arithmetic structure of the numbers
- In digital logic we only work with two states – HIGH and LOW.
- It is clumsy to work with numbers between 0 – 9 (decimal) to perform mathematics with a two state (binary) system.
- We use different number systems to make life easier when working with and designing digital systems.

Decimal numbers

- Just as you (probably) have ten fingers, there are ten decimal digits from 0 to 9. [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
- Each digit (right to left) represents a multiple of a power of 10.
- Decimal numbers are also called **base-10** (or radix-10) numbers.

max 9

1000's	100's	10's	1's
9	7	4	2

Indicating the **base** of the number



$$9742_{10} = 9 \times 10^3 + 7 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

nine thousandsseven hundredsfour tenstwo ones

- An N -digit decimal number represents one of 10^N possibilities. This is called the **range** of the number.
 - Three-digit decimal number represents one of 1000 possibilities in the range of 0 to 999.

Binary numbers

- Binary digits or “bits” represent one of two values, 0 or 1.
- Each bit (right to left) represent a multiple of a power of 2.
- Binary numbers are **base-2**. *max 15*

16's	8's	4's	2's	1's
1	0	1	1	0

$$10110_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 22_{10}$$

one
sixteenno
eightone
fourone
twono
one

- An N -bit binary number represents one of 2^N possibilities.
 - A four-bit binary number represents one of 16 possibilities. For positive numbers, this is the range of 0 to 15.
- Binary numbers are convenient for logic circuits in computers, since they can represent 2 states, i.e. low and high voltages.

Number interpretation

- The interpretation of a number depends on the base number used:

Decimal: 1101 $= 1 \times 10^3 + 1 \times 10^2 + 0 \times 10^1 + 1 \times 10^0$
 $= 1000 + 100 + 1$
 $= \text{one thousand, one hundred and one}$

Binary: 1101 $= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
 $= 8 + 4 + 1$
 $= 13$
 $= \text{thirteen}$

- The base number, or radix, is indicated with a subscript when there is potential for confusion:

$$(1101)_2 = (13)_{10}$$

Conversion between Decimal and Binary

Convert 84_{10} to binary $\rightarrow$ go to base 2

Solution 1 (Left to Right)

Find largest power of 2 less than or equal to the number.
Subtract and repeat until 0.

2 ^{biggest number} $1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

Answer

$1010100_2 = 20$

$20 - 16$

4

$4 - 4 = 0$

3 $\Rightarrow 1010100$

$1_{2^6} 0_{2^5} 1_{2^4} 0_{2^3} 1_{2^2} 0_{2^1} 0_{2^0} \Rightarrow$ sum together to get back to decimal

1 $2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

Conversion between Decimal and Binary

Convert 84_{10} to binary

Solution 2 (Right to Left)

Repeatedly divide the number by 2, the remainder goes in each column.

Answer

1010100_2

Hexadecimal numbers max 15

- Decimal numbers are useful, because they are used by people.
- Binary numbers are useful, because they are used by computers.
- Hexadecimal (**base-16**) numbers are useful, because they provide a shorthand notation for binary numbers.
- Each hexadecimal digit represents a group of four bits (a nibble).
- Hexadecimal numbers use the **digits 0 to 9** along with the **letters A to F**, to represent a total of 16 possible values (from 0 to 15).
- Each digit (right to left) represents a multiple of a power of 16.

256's	16's	1's
2	E	D

→

$$2ED_{16} = 2 \times 16^2 + E \times 16^1 + D \times 16^0 = 749_{10}$$

two
two hundred
fifty six's fourteen
sixteens thirteen
ones

Conversion table

Table 1.2 Hexadecimal number system

Hexadecimal Digit	Decimal Equivalent	Binary Equivalent
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Binary to hexadecimal conversion

- A binary number is converted to a hexadecimal number by taking groups of four bits, starting from the bit in the 1's column, and replacing them with the corresponding hexadecimal digits.
- A 16-bit binary number can be represented with four hex digits, for example:

$$(1010111100100101)_2 = (AF25)_{16}$$

because 1010 1111 0010 0101

A F 2 5



- If the number of bits is not a multiple of four, then add 0's to the left of the binary number.
- Conversion from hexadecimal to binary is also straightforward: replace each hexadecimal digit by four bits that represent the same value

Conversion between Decimal and Hexadecimal

Convert 333_{10} to hexadecimal

Solution 1 (Left to Right)

Find largest power of 16 less than or equal to the number.
Subtract and repeat until 0.

Answer $1 \times 16^2 + 4 \times 16^1 + D \times 16^0$

$$14D_{16} \quad \begin{array}{l} 333 - 256 \\ = 77 \end{array} \quad \begin{array}{l} 77 - 64 \\ = 13 \end{array} \quad \Rightarrow 14D$$

$$\begin{aligned} 16^0 &= 1 \\ 16^1 &= 16 \\ 16^2 &= 256 \\ &\dots \end{aligned}$$

Conversion between Decimal and Hexadecimal

Convert 333_{10} to hexadecimal

Solution 2 (Right to Left)

Repeatedly divide the number by 16, the remainder goes in each column.

Answer

$14D_{16}$

Octal numbers

- Octal (**base-8**) numbers uses the **digits from 0 to 7**. A less used alternative to writing binary in shorthand.
- A 12-bit binary number can be represented with four octal digits, for example:

$$(111100100101)_2 = (7445)_8$$

because 111 100 100 101

7 4 4 5

- Why do engineers confuse Halloween with Christmas?
⇒ 31 OCT = 25 DEC

Bytes, Nibbles, LSB, MSB

bit
1 0 1 1 0 0 1

- Each digit in a binary number is called a **bit**.
- A group of *eight bits* is called a **byte**. ($2^8=256$ possible values)
- A group of *four bits* is called a **nibble**. ($2^4=16$ possible values)
- Microprocessors handle data in chunks called **words**. The size of a word depends on the architecture of the microprocessor.
 - E.g. 64-bit processor operates in 64-bit words.
- The bit in the 1's column is called the **least-significant bit (lsb)**
- The bit on the other end is called the **most-significant bit (msb)**
- Similarly, within a word, the bytes are identified as the **least significant byte (LSB)** through **most significant byte (MSB)**

101100
most least
significant significant
bit bit

(a)

DEAFDAD8
most least
significant significant
byte byte

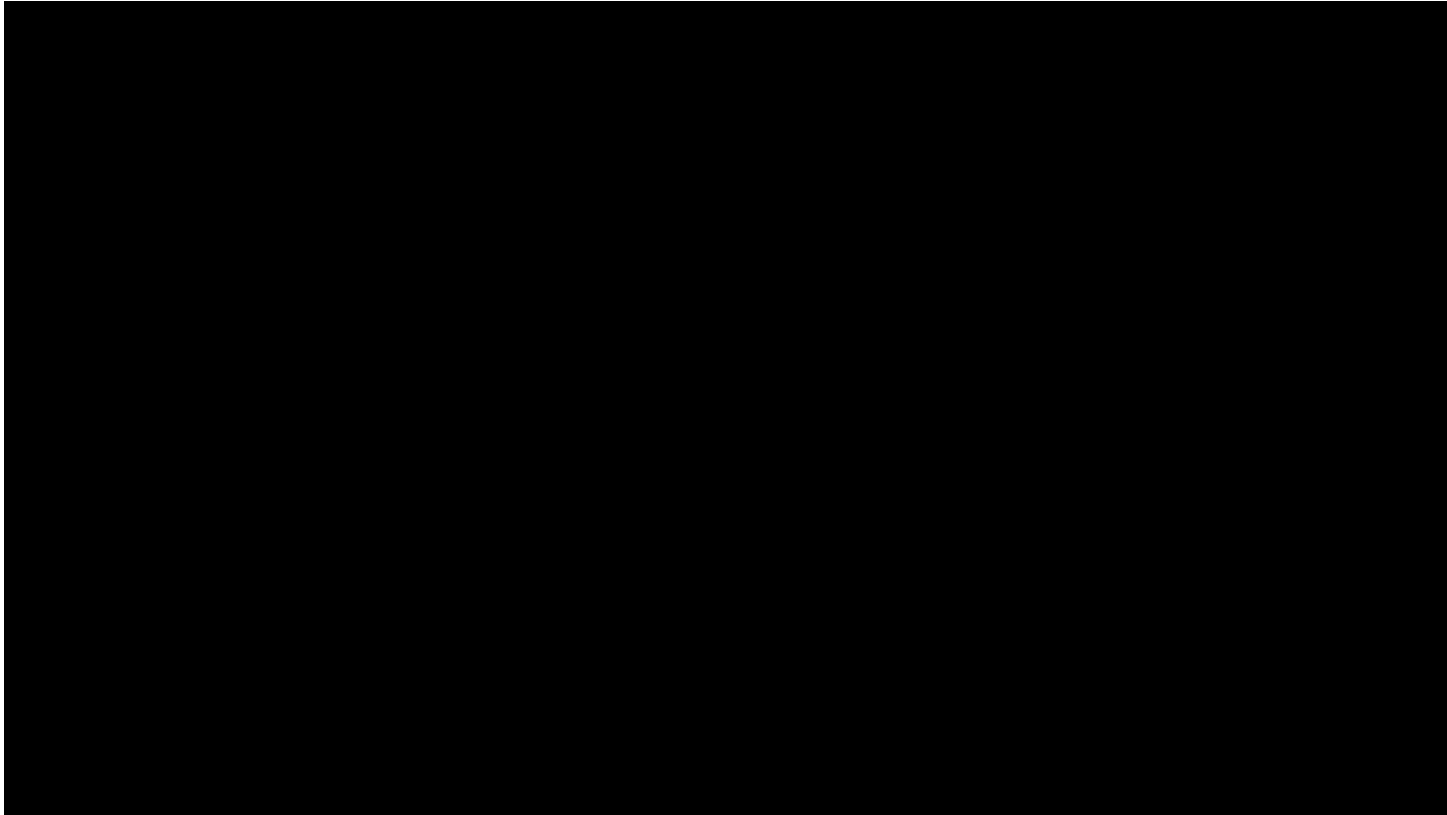
(b)

NB

Number sizes

- By handy coincidence:
 - $2^{10} = 1024 \approx 10^3 \Rightarrow$ **kilo (upper case K to disambiguate from k =1000)**
 - $2^{20} = 1048576 \approx 10^6 \Rightarrow$ **mega**
 - $2^{30} = 1073741824 \approx 10^9 \Rightarrow$ **giga**
- 1024 bytes is called a **kilobyte (KB)**
- 1024 bits is called a **kilobit (Kb)** (note lowercase b)
- Similarly MB, Mb, GB, Gb...

Representing letters (ASCII)



Source: <https://www.youtube.com/watch?v=1GSjbWt0c9M>

ASCII table

Table 6.5 ASCII encodings

#	Char	#	Char	#	Char	#	Char	#	Char	#	Char
20	space	30	0	40	@	50	P	60	`	70	p
21	!	31	1	41	A	51	Q	61	a	71	q
22	"	32	2	42	B	52	R	62	b	72	r
23	#	33	3	43	C	53	S	63	c	73	s
24	\$	34	4	44	D	54	T	64	d	74	t
25	%	35	5	45	E	55	U	65	e	75	u
26	&	36	6	46	F	56	V	66	f	76	v
27	'	37	7	47	G	57	W	67	g	77	w
28	(	38	8	48	H	58	X	68	h	78	x
29	)	39	9	49	I	59	Y	69	i	79	y
2A	*	3A	:	4A	J	5A	Z	6A	j	7A	z
2B	+	3B	;	4B	K	5B	[	6B	k	7B	{
2C	,	3C	<	4C	L	5C	\	6C	l	7C	
2D	-	3D	=	4D	M	5D	]	6D	m	7D	}
2E	.	3E	>	4E	N	5E	^	6E	n	7E	~
2F	/	3F	?	4F	O	5F	_	6F	o		

Declaring/Printing C ints in Different Number Systems

- In C the variable type `int` is stored as binary number of at least 16 bits
 - Declaration and presentation may be expressed in higher-base number systems, with varying syntax
 - Declaration:
 - `int x = 11;` declares `x` as the number 11 in base-10 (decimal)
 - `int x = 0b1011;` declares `x` as the same value in base-2 (binary)
 - `int x = 013;` declares `x` as the same value in base-8 (octal)
 - `int x = 0xB;` declares `x` as the same value in base-16 (hexadecimal)
 - Presentation:
 - `printf("%d", x);` prints `x` in base-10 (decimal)
 - `printf("%o", x);` prints `x` in base-8 (octal)
 - `printf("%x", x);` prints `x` in base-16 (hexadecimal)
- NB exam Q to code*
- NB: There is no standardised C syntax to print in base-2 (binary)

Declaring/Printing C ints in Different Number Systems

```
1  #include <stdio.h> /* printf */
2  #include "dtob.h" /* decimal_to_binary */
3
4  int main()
5  {
6      // Decimal number
7      int x = 17;
8      printf("Decimal: %td\n", x);
9      printf("Binary: %ts\n", decimal_to_binary(x));
10     printf("Hexadecimal: %tx\n\n", x);
11
12     // Binary number
13     int y = 0b1011;
14     printf("Decimal: %td\n", y);
15     printf("Binary: %ts\n", decimal_to_binary(y));
16     printf("Hexadecimal: %tx\n\n", y);
17
18     // Hexadecimal number
19     int z = 0xbf;
20     printf("Decimal: %td\n", z);
21     printf("Binary: %ts\n", decimal_to_binary(z));
22     printf("Hexadecimal: %tx\n\n", z);
23
24     return 0;
25 }
```

```
Decimal:      17
Binary:       00010001
Hexadecimal:  11
```

```
Decimal:      11
Binary:       00001011
Hexadecimal:  b
```

```
Decimal:      191
Binary:       10111111
Hexadecimal:  bf
```

Next lecture...

- Signed numbers (positive and negative numbers)
- Addition and subtraction of binary numbers

Concluding remark

There are 10 kinds of people in the world: Those who understand binary and those who don't.